TRANSCRIPTION

Defined as the process of synthesizing RNA using DNA has template.

TRANCRIPTION UNIT:

- A transcription unit is defined as that region of DNA that includes the signals for transcription initiation, elongation, and termination.
- DNA-dependent RNA polymerase is the enzyme responsible for the polymerization of ribonucleotides into a sequence complementary to the template strand of the gene.
- The enzyme attaches at a specific site –**the promoter**- on the template strand.
- This is followed by initiation of RNA synthesis at the starting point, and the process continues until a termination sequence is reached.

BACTERIAL DNA DEPENDENT RNA POLYMERASE

The DNA dependent RNA polymerase (RNAP) of the bacterium Escherichia coli exists as an approximately 400kDa core complex consisting of two identical α subunits, similar but not identical β subunits, an ω subunit and a sigma (σ). Beta is thought to be catalytic subunit.

RNAP, a metalloenzyme, also contains two zinc molecules. The core RNA polymerase associates with the specific proteins factor (the sigma factor) that helps the core enzyme recognize and bind to specific deoxynucleotide sequence of the promoter region to form the pre-initiation complex (PIC). Bacteria contains multiple factors, each of which acts as a regulatory protein.

PROKARYOTIC TRANSCRIPTION:

Steps of RNA synthesis- The process of transcription of a typical gene of *E.coli* can be divided into three phases-

i. Initiation
ii. Elongation
iii. Termination

Initiation of Transcription:

Initiation of transcription involves the binding of the RNA polymerase holoenzyme to the promoter region on the DNA to form a preinitiation complex or PIC. Characteristic "Consensus" nucleotide sequence of the prokaryotic promoter region are highly conserved.

Pribnow box: This is a stretch of 6 nucleotides(5`-TATAAT-3`) centered about 8-10 nucleotides to the left of the transcription start site.

35 sequence: A second consensus nucleotide sequence(5`-TTGACA-3`), is centered about 35 bases to the left of the transcription start site.

Binding of RNA polymerase (RNAP) to the promoter region is followed by a conformational change of the RNAP, and the first nucleotide (almost always a purine) then associates with the initiation site on the subunit of the enzyme.

In the presence of the appropriate nucleotide, RNAP catalyzes the formation of a phosphodiester bond, and the nascent chain is now attached to the polymerization site on the subunit of RNAP.

In both prokaryotes and eukaryotes, a purine ribonucleotide is usually the first to be polymerized into the RNA molecule.

After 10-20 nucleotides have been polymerized, RNAP undergoes a second conformational change leading to promoter clearance.

Once this transition occurs, RNAP physically moves away from the promoter, transcribing down the transcription unit, leading to the next phase of the process, elongation.

Elongation step of Transcription

- → As the elongation complex containing the core RNA polymerase progresses along the DNA molecule, DNA unwinding must occur in order to provide access for the appropriate base pairing to the nucleotides of the template strand.
- → The extent of this transcription bubble (i.e., DNA unwinding) is constant throughout and is about 20 base pairs per polymerase molecule.
- → RNA polymerase has associated with it an "unwindase" activity that opens the DNA helix.
- → Topoisomerase both precedes and follows the progressing RNAP to prevent the formation of super helical complexes.
- → Base pairing rule is followed during the incorporation of ribonucleotides.

Termination of transcription

Termination of the synthesis of the RNA molecule in bacteria is of two types

(A)Rho (ρ) dependent termination-

- The termination process is signalled by a sequence in the template strand of the DNA molecule- a signal that is recognized by a termination protein, the rho (ρ) factor.
- Rho is an ATP dependent RNA stimulated helicase that disrupts the nascent RNA-DNA complex.

(B)Rho independent termination

- This process requires the presence of intrachain self-complementary sequences in the newly formed primary transcript so that it can acquire a stable hair pin turn that slows down the progress of the RNA polymerase and causes it to pause temporarily.
- Near the stem of the hairpin, a sequence occurs that is rich in G and C.
- This stabilizes the secondary structure of the hairpin.
- Beyond the hair pin, the RNA transcript contains a strings of `U`s, the bonding of `U`s to the corresponding `A`s is weak.
- This facilitates the dissociation of the primarily transcript from DNA.
- After termination of the synthesis of the RNA molecule, the enzyme separates from the DNA template.
- With the assistance of another factor, the core enzymes then recognizes a promoter at which the synthesis of a new RNA molecule commences.

EUKARYOTIC TRANSCRIPTION

- ➔ The general process of transcription can be applied to both prokaryotic cells and eukaryotic cells.
- ➔ The basic biochemistry for each is the same; however, the specific mechanisms and regulation of transcription differ between prokaryotes and eukaryotes.
- ➔ Transcription of eukaryotic genes is far more a complicated process than prokaryotes.

1) Location
   a) In eukaryotes, transcription occurs in the cell's nucleus, mRNA then moves to the cytoplasm for translation.
2) Genome size
   a) The genome size is much larger in eukaryotes.
   b) Greater specificity is needed for the transcription of eukaryotic genes.
3) Chromatin Structure
   a) Eukaryotic DNA is wrapped around proteins called histones to form structures called nucleosomes. Eukaryotic DNA is packed to form chromatin. While RNA polymerase interacts directly with prokaryotic DNA, other proteins mediate the interaction between RNA polymerase and DNA in eukaryotes.
4) RNA polymerases
   a) There are three distinct classes of RNA polymerases in eukaryotic cells. All are large enzymes with multiple subunits. Each class of RNA polymerase recognises particular types of genes.
   b) RNA polymerase I- Synthesizes the precursor of the large ribosomal RNAs(28S, 18S and 5.8S).
   c) RNA polymerase II – Synthesizes the precursors of messenger RNA and small nuclear RNAs(snRNAs).
   d) RNA polymerase III – Synthesizes small RNA, including t RNAs, small 5S RNA and some snRNAs.
5) Promoter regions
   a) Eukaryotic promoters are more complex.
   b) Two types of sequence elements are promoter proximal and disyal regulatory elements.
   c) There are two elements in promoter proximal, one of these defines where transcription is to commence along the DNA , and the other contributes to the mechanisms that control how frequently this event is to occur.
   d) Most mammalian genes have a TATA box that is usually located 25-30 bp upstream from the transcription start site.
   e) Typical of these DNA elements are the GC and CAAT boxes, so named because of the DNA sequences involved.
   f) Each of these boxes binds a specific protein.
   g) Distal regulatory elements enhance or decrease the rate of transcription.
   h) They include the enhancer / silencer regions and other regulatory elements.
6) Promoter identification
   a) In contrast to the situation in prokaryotes, eukaryotic RNA polymerase alone are not able to discriminate between promoter sequences and other regions of DNA.

b) The TATA box is bound by 34 kDa TATA binding protein (TBP), which in turn binds several other proteins called TBP-associated factors (TAFs).
c) This complex of TBP and TAFs is referred to as TFIID.
d) Formation of the basal transcription complex begins when TFIID binds to the TATA box.
e) It directs the assembly of several other components by protein-DNA and protein-protein interactions.
f) The entire complex spans DNA from position -30 to +30 relative to the initiation site.
g) Binding of TFIID to the TATA box sequence is thought to represent the first step in the formation of the transcription complex on the promoter.
h) Another set of proteins co-activators help regulate the rate of transcription initiation by interacting with transcription activators that bind to upstream DNA elements.

7) Enhancers and Repressors
   a) A third class of sequence elements can either increase or decrease the rate of transcription initiation of eukaryotic genes.
   b) These elements are called either enhancers or repressors(or silencers), depending on which effect they have.
   c) They have been found in a variety of locations both upstream and downstream of the transcription start site and even within the transcribed portions of some genes.
   d) In contrast to proximal and upstream promoter elements, enhancers and silencers can exert their effects when located hundreds, or even thousands of bases away from transcription units located on the same chromosome.

8) Termination of transcription
   a) The signals for the termination of transcription by eukaryotic RNA polymerase II are very poorly understood.

9) Processing of primary transcript
   a) mRNA produced as a result of transcription is not modifies in prokaryotic cells. Eukaryotic cells modify mRNA by RNA splicing, 5' end capping and addition of a polyA tail.
   b) Most eukaryotic RNAs are synthesized as precursors that contain excess sequences which are removed prior to the generation of mature, functional RNA.

POST TRANSCRIPTIONAL MODIFICATIONS

- Primary transcript made by RNA polymerase normally undergo further alteration, called **post transcriptional processing or modifications**.
- Prokaryotes: mRNA transcribed directly from DNA template and used immediately in protein synthesis.
- Eukaryotes: primary transcript (hnRNA) must be processed to produce the mRNA(active form).
- It occurs in nucleus of cell.

POST TRANSCRIPTIONAL MODIFICATIONS OF RNA ACCOMPLISH THESE THINGS

1. Modifications help the RNA molecule to be **recognized by molecules** that mediate RNA translation into proteins.
2. During post-transcriptional processing, **portions of the RNA chain that are not supposed to be translated into proteins are cut out of the sequence**.
3. Without post-transcriptional processing, **protein synthesis could be significantly slowed**, since it would take longer for translation machinery to recognize RNA molecules.

Post transcriptional modification mainly includes

- 5' capping
- Addition of poly-A tail
- Splicing

5' END CAPPING

- At the end of transcription, the 5' end of the RNA transcript contains a free triphosphate group since it was the first incorporated nucleotide in the chain. The capping process replaces the triphosphate group with another structure called the "cap". The cap is added by the enzyme guanyl transferase. This enzyme catalyzes the reaction between the 5' end of the RNA transcript and a guanine triphosphate (GTP) molecule.
- In the reaction, the beta phosphate of the RNA transcript displaces a pyrophosphate group at the 5' position of the GTP molecule. The cap is formed through a 5'-5' linkage between the two substrates.
- Capping protects the 5' from enzymatic degradation in the nucleus and assists in export to the cytosol. Eukaryotic mRNAs lacking the cap are not efficiently translated.

The Poly A Tail

- Post-transcriptional RNA processing at the opposite end of the transcript comes in the form of a **string of adenine bases** attached to the end of the synthesized RNA chain.
- The addition of the adenines is catalyzed by the enzyme poly A polymerase.
- The mRNA is first cleaved about 20 nucleotides downstream from an AAUAA recognition sequence.
- Another enzyme, poly A tail appears to protect the 3` end of RNA from 3` 5` nuclease attack.

- Histone and interferon`s mRNAs lack poly A tail.
- After the mRNA enters the cytosol, the poly A tail is gradually shortened.

SPLICING

- **Removal of introns (Splicing)**
- Introns or intervening sequences are the RNA sequences which do not code for the proteins.
- These introns are removed from the primary transcript in the nucleus, exons (coding sequences) are ligated to form the mRNA molecule and the mRNA molecule is transported to the cytoplasm.
- The molecular machine that accomplishes the task of splicing is known as the **spliceosome**.
- Small nuclear RNA molecules that recognize splice sites in the presence mRNA sequence.
- The excised intron is released as a "lariat" structure, which is degraded

INHIBITORS OF TRANSCRIPTION

➔ Rifampicin- binds with Beta subunit of prokaryotic RNA polymerase. It is an inhibitor of prokaryotic transcription initiation. It binds only to bacterial RNA polymerase but not to eukaryotic RNA polymerases. Therefore, Rifampicin is a powerful drug for treatment of bacterial infections. Used for the treatment of tuberculosis and leprosy.

➔ Actinomycin D-Incalates with DNA strands. Actinomycins inhibit both DNA synthesis and RNA synthesis but blocking chain elongation. They interact with G-c base pairs as they require the 2-amino group of guanine for binding. Actinomycins are used as anti-cancer drugs.

➔ Mitomycin: Mitomycin intercalates with DNA strands, blocks transcription hence used as anticancer drug.

➔ Alpha amanitin: Alpha amanitin is a molecule made from the "death cap" mushroom and is a known potent inhibitor RNA polymerase. One single mushroom could very easily lead to a fast death in 10 days. The mechanism of action is that alpha amanitin inhibits RNA polymerase-II at both the initiation and elongation states of transcription.